gpeyt1-14 · Moisture conditioning

GEO PE · Long Calculation Problems I — Topic 3: Construction Observation, QA/QC & Safety

Problem

Borrow for a 52,000 ft³ (compacted) highway embankment has an in-situ water content of 7.5%. The specification requires placement at 13.0% and a dry unit weight of 114 pcf. The water truck holds 4,000 gal. The borrow's in-place moist unit weight is 118 pcf and Gs = 2.68 (not needed).

Most nearly, the volume of water that must be added is:

A. 39,140 gal
B. 92,514 gal
C. 5,225 gal
D. 40,514 gal

Given equations / data

> Gallons = γd,desired × Vcompacted × (wdesired − wborrow)/100 / 8.33 lb/gal

Worked solution

1. Δw = 13.0 − 7.5 = 5.5%
2. Water weight = 114 × 52,000 × 0.055 = 326,040 lb
3. Gallons = /8.33 = 39,140 gal (≈ 10 truckloads)

Final answer

39,140 gal

Why the other values are wrong

- **92,514 gal** — used the full desired w (ignored borrow moisture)
- **5,225 gal** — divided by 62.4 (got ft³, labeled gal)
- **40,514 gal** — used the moist borrow unit weight

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §2.3.1 Earthwork/compaction — amount of water to add (Ch. 2 Construction)

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=v_1tQfxBowQ, https://www.youtube.com/watch?v=f02U9lj1I58 (no creator values copied).

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